

Total No. of Questions :10]

SEAT No. :

P3458

[5670]-734

[Total No. of Pages :2

B.E. (IT)

DISTRIBUTED COMPUTING SYSTEM

(2015 Course) (414462) (Semester-II) (End Sem.)

Time : 2½Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2 and Q.3 or Q.4 and Q.5 or Q.6 and Q.7 or Q.8 and Q.9 or Q.10,
- 2) Figures to the right indicate full marks.
- 3) Neat diagrams must be drawn wherever necessary.
- 4) Assume suitable data, if necessary.

Q1) a) Outline the goals of distributed systems. **[4]**

b) Explain with example different system architectures of distributed system. **[6]**

OR

Q2) a) Point out various scalability issues and the techniques to resolve them while designing distributed system. **[6]**

b) Explain Bully algorithm for coordinator election. **[4]**

Q3) a) Explain centralized mutual exclusion algorithm. List the shortcomings of this algorithm. **[6]**

b) Discuss the solution proposed by vasudevan et al. that can handle failing nodes and partitioning networks while electing a coordinator in wireless environments. **[4]**

OR

Q4) a) Classify the different types of failures based on their severity. Explain the behaviour of the server in each case. **[6]**

b) Explain how reliable group communication can be achieved in distributed systems. **[4]**

Q5) a) Illustrate with an example the working of Bit Torrent protocol. **[8]**

b) List and explain the design goals of the sun NFS. **[8]**

OR

P.T.O.

- Q6)** a) Explain the design of distributed file system architecture w.r.t HDFS.[8]
b) Explain the following terms w.r.t naming entities. [8]
i) Names ii) Identifiers
iii) Name spaces iv) Addresses

- Q7)** a) Draw and explain the principle of providing and using a web service.[8]
b) Give the drawback of using hierarchical caches for a web proxy. Discuss the solution to overcome the drawback. [8]

OR

- Q8)** a) Discuss the general organization of Apache web server. Explain the concept of a hook. [8]
b) Why web services are used? Explain SOAP and REST based web services. [8]

- Q9)** a) Explain key management for secret and public key distribution with suitable block diagram. [9]
b) List out the various security threats in a distributed system. Elaborate on the various policies and security mechanisms in distributed systems. [9]

OR

- Q10)** a) Discuss how Authorization management helps in maintaining the security in a distributed system. [9]
b) With suitable diagram, Explain the various configurations of firewall. [9]

